

OnionMonad: Onion Circuits as Client-Sealed Loops of Registered Operation Symbols

Ryan J. Kung

Abstract

The Rings onion data plane interprets two layers, relay and exit, and answers along the reversed path through relays that keep per-circuit return state and forward one client-bound ciphertext byte-identical. We generalise the circuit to a term over a static signature Σ of operation symbols: nodes are partial Σ -algebras that register symbols, never applications; a circuit is a client-sealed loop $\diamond \rightarrow g \rightarrow \dots \rightarrow g \rightarrow \diamond$ that evaluates one application per hop and composes them in the Kleisli category of the hop effect monad, with **relay** as the identity. Each segment holds $s = 2$ relays, the guard counting in both end segments; every hop parses one uniform layer of a fixed-length Sphinx-style header; values travel in a fixed-width carry slot enciphered with AEZ, one length-preserving layer per relay and one authenticated layer per consumer; replies to world-facing sessions ride single-use reply blocks under credit flow control. We state eleven laws, among them noninterference over hop views, position hiding, client path authority (which admits Arrow composition and excludes monadic bind), byte unlinkability, and single-use admission, list the residual leakage, and map each law to a test obligation.

I. Introduction

Today's circuit is the term **relay**^k $\ggg (svc, body)$ evaluated on the forward path and answered on its reverse (§X). This paper states the general structure of which it is the degenerate instance, as definitions D1–D9 and laws L1–L10 and L5' of issue #834, numbered identically. Design lineage: Sphinx headers and wide-block payloads [1], reply blocks [2], [3], loops [4], AEZ [5], Kleisli semantics [6], generic effects [7], [8], and the Arrow/Selective hierarchy [9]–[11].

II. Signature and Algebras

Definition D1 (signature). Σ is a finite, statically defined table of symbols; each $f \in \Sigma$ has

$$\text{spec}(f) = (\text{name}, \text{Args}_f, \text{In}_f, \text{Out}_f, \\ W(f), L(f), \text{pos}(f), \text{quota}(f))$$

with *name* identifying the backend (e.g. `jev.typesafe`), Args_f client constants, $\text{In}_f, \text{Out}_f$ carried types, $W(f) \in \mathbb{N}$ the exact output width in bytes, $L(f)$ the exact latency class, $\text{pos}(f) \subseteq \{\text{Intermediate}, \text{WorldFacing}\}$, and $\text{quota}(f)$ the admission budget. W, L are protocol data. The identity symbol is

$$\text{relay} : X \rightarrow X, \quad L(\text{relay}) = 0;$$

relay has no W : it preserves the slot width C (L3). Σ is closed: a name outside Σ is rejected at configuration,

Ryan J. Kung can be reached at ryankung@ieee.org. Manuscript created September 24, 2026. This document freezes the design of Rings issue #834 (rings-node, module onion) before its Phase 2 code; it is a specification, not shipped behaviour. The constants $s = 2$ and $n_{\max} = 4$ are those fixed in the issue discussion.

at descriptor decode, and at route construction; nothing falls back to tcp, and operator-named services are not accepted. An operator exposing TCP targets registers tcp and restricts targets by policy.

Definition D2 (registration). Node n is a partial Σ -algebra: a set $\Sigma_n \subseteq \Sigma$ and, for each $f \in \Sigma_n$, a generic effect [7]

$$[[f]]_n : \text{Args}_f \times \text{In}_f \rightarrow M \text{Out}_f,$$

M the hop effect monad (failure, delay, world I/O). n publishes $f \in \Sigma_n$, never an application. Every published symbol, **relay** included, carries n 's process epoch e_n , drawn afresh at each process start: **relay** in `OnlineNodeDescriptor::capabilities` extended with e_n , every other f in one signed descriptor per symbol with e_n and $\text{quota}(f)$. Registration includes **relay**: $\Sigma_n \neq \emptyset \Rightarrow \text{relay} \in \Sigma_n$, so a neighbour's registry entry does not reveal whether a relay is adjacent to a symbol hop.

A node is a handler [8] for the operations it registers; **WorldFacing** symbols (tcp, https) may hold a session, **Intermediate** symbols are one-shot.

Definition D3 (application). An application is $(f, \bar{a})$, $\bar{a} \in \text{Args}_f$, typed $(f, \bar{a}) : \text{In}_f \rightsquigarrow \text{Out}_f$; it is sealed in the layer of the hop that evaluates it and visible to no other hop.

III. Pipelines

Definition D4a (pipeline). Pipelines are the morphisms of the free category $\mathcal{F}(\Sigma)$ on the graph with types as vertices and applications as edges; composition is

$$\frac{\theta : X \rightsquigarrow Y \quad \vartheta : Y \rightsquigarrow Z}{\theta \ggg \vartheta : X \rightsquigarrow Z}.$$

$P = (f_1, \bar{a}_1) \ggg \dots \ggg (f_n, \bar{a}_n)$ is well formed iff

$$\begin{aligned} 1 \leq n \leq n_{\max} = 4, \quad \forall k < n : \text{Out}_{f_k} = \text{In}_{f_{k+1}}, \\ \forall k < n : \text{Intermediate} \in \text{pos}(f_k), \\ f_n \text{ is a WorldFacing session symbol} \Rightarrow n = 1. \end{aligned}$$

Violations are rejected at construction. The bound $n_{\max}$ is MAX_ONION_LOOP_SYMBOLS.

Definition D4b (Kleisli denotation). For $\phi : X \rightarrow MY$, $\psi : Y \rightarrow MZ$ in $\mathbf{KI}(M)$ [6], $\phi \ggg \psi = \mu_Z \circ M\psi \circ \phi$. For hop assignment $h_1..h_n$ with $f_k \in \Sigma_{h_k}$,

$$\llbracket P \rrbracket(v_0) = (\llbracket f_1 \rrbracket_{h_1}(\bar{a}_1, -) \ggg \dots \ggg \llbracket f_n \rrbracket_{h_n}(\bar{a}_n, -))(v_0),$$

the unique functor $\mathcal{F}(\Sigma) \rightarrow \mathbf{KI}(M)$ extending the hop interpretations. The client consumes $\llbracket P \rrbracket(v_0)$.

Law L1 (identity). $\llbracket \text{relay} \rrbracket = \eta$, hence $\text{relay} \ggg f = f = f \ggg \text{relay}$ on values. The loop builder factors through nf , which deletes relay applications: $\text{Loop}(P) = \text{Loop}(\text{nf}P)$, so $\text{Loop}(\text{relay} \ggg P) \stackrel{d}{=} \text{Loop}(P)$ and every hop's view is unchanged.

Law L2 (associativity). $(f \ggg g) \ggg h = f \ggg (g \ggg h)$ on values; immediate from the monad laws, and $\mathcal{F}(\Sigma)$ has no bracketing.

nf and L1 on views apply from Phase 2a. Phase 1 admits only $\text{relay}^k \ggg (f, \bar{a})$ with $\text{WorldFacing} \in \text{pos}(f)$, which is today's circuit (§X).

IV. The Loop

Definition D4c, D5 (loop, segment). Let $\diamond$ be the client and g its entry guard. Segment $k \in [0, n]$ is $s = 2$ relays $r_{k,1}..r_{k,s}$ (ONION_SEGMENT_RELAYS) from producer p_k to consumer c_k ,

$$\begin{aligned} p_0 = \diamond, \quad p_k = h_k, \quad c_k = h_{k+1}, \quad c_n = \diamond, \\ \text{Loop}(P) : \diamond \rightarrow r_{0,1}..s \rightarrow h_1 \rightarrow r_{1,1}..s \rightarrow \dots \\ \rightarrow h_n \rightarrow r_{n,1}..s \rightarrow \diamond. \end{aligned}$$

Positions are $i \in [1, H]$; position i evaluates f_i , which is relay on every $r_{k,j}$. $\text{Loop}(\text{nf}P)$ is the symbol word $\text{relay}^s f_1 \text{relay}^s \dots f_n \text{relay}^s$.

Law L7 (guard closure). $r_{0,1} = r_{n,s} = g$: the guard is relay 1 of segment 0 and relay s of segment n . The other $H - 2$ positions carry pairwise distinct DIDs, distinct from g . g is the only hop adjacent to $\diamond$.

Proposition 1 (Loop size). With n symbol hops and $n + 1$ segments of s relays,

$$\begin{aligned} H(n, s) &= n + (n + 1)s = (s + 1)n + s, \\ \# \text{DID} &= H - 1, \quad E = H + 1 \text{ cells}, \\ \hat{H} &= H(n_{\max}, s) = (s + 1) \cdot 4 + s = 14. \end{aligned}$$

Proof. Segments contribute $(n + 1)s$ positions; L7 repeats exactly one DID; the H positions lie on a path from $\diamond$ to $\diamond$ of $H + 1$ edges. $\square$

Table I
Loop shape at $s = 2$ (Prop. 1)

n	H	DIDs	E	header $ \beta $
1	5	4	6	$\hat{H}\ell = 9086 \text{ B}$
2	8	7	9	
3	11	10	12	
4	14	13	15	

Proposition 2 (Segment length). $s = 2$ is the least s at which no single node is adjacent to both $\diamond$ and a symbol hop.

Proof. $s = 0$ violates L7. At $s = 1$, $r_{0,1} = r_{0,s} = r_{n,1} = r_{n,s} = g$, so g alone is adjacent to $\diamond$, h_1 and h_n , and for $n = 1$ links $\diamond$ to its world-facing h_1 . At $s = 2$, g neighbours only $\diamond$, $r_{0,2}$, $r_{n,1}$. $\square$

Relays do not mix ($L(\text{relay}) = 0$), so a timing-correlating coalition skips middle relays: $s \geq 3$ adds $n + 1$ edges of latency per relay and no anonymity. At $(s, n) = (2, 1)$, $E = 6$ and the forward prefix $g, r_{0,2}, h_1$ is Tor's three-hop circuit [12].

Route validation checks $H = (s + 1)n + s$ with exactly s relays between consecutive symbol hops, and admits g exactly twice, at positions 1 and H (has_duplicate_dids); selection fails closed when fewer than $H - 1$ distinct eligible DIDs exist. MAX_ONION_LOOP_HOPS = $\hat{H}$ is derived from s and $n_{\max}$ and replaces MAX_ONION_CIRCUIT_HOPS = 8; DEFAULT_ONION_ROUTE_HOPS and OnionRouteRequest::{hop_count, allow_short_paths} are removed, since a short path is a distinguishable segment length (D5).

V. Uniform Layer and Timing

Definition D6 (uniform layer). Every position i has one layer shape

$$\lambda_i = (f_i, \bar{a}_i, \text{next}_i, e_i, x, \nu_i, \kappa_i, \gamma_{i+1}, \alpha_{i+1})$$

of ℓ bytes, with $\bar{a}_i$ padded to A bytes: epoch e_i , one expiry per loop,

$$\begin{aligned} x &= \lceil t_{\text{build}}/Q \rceil Q + X_0, \\ Q &= 30 \text{ s}, \quad X_0 = V - Q = 120 \text{ s}, \end{aligned}$$

a function of the build quantum only (X_0 is today's ONION_FORWARD_PAYLOAD_TTL_MS), replay nonce ν_i , carry keys $\kappa_i = (k_i^{\text{in}}, k_{i,1..s+1}^{\text{out}})$, and the next hop's MAC γ_{i+1} and key transport α_{i+1} (Def. 1). A relay has $\bar{a}_i = ()$ and random k^{out} . Each loop has one class $b \in \mathfrak{B}$, today's cell buckets above 4 KiB (16 KiB to 12 MiB), chosen by $\diamond$; the cell on every edge of the loop, SURB replies included, is

$$(\chi_{i+1}, y_i), \quad \chi = (\alpha, \beta, \gamma), \quad |\beta| = \hat{H}\ell, \quad |y_i| = C_b,$$

padded to b , with $C_b = b - |\chi| - F$ and F the cell framing.

With today's encodings (239-byte ElGamal-AEAD envelope as α , 43-byte DIDs, 52-byte public keys) and

48-byte AEZ keys, $\ell(s, A) = 393 + A + 48(s + 2)$, so $\ell(2, 64) = 649$ B and $|\beta| = 9086$ B: every loop cell uses at least the 16 KiB bucket.

Definition 1 (Fixed-length header). Hop i derives K_i from α_i and its secret key and expands $\rho_i = \text{PRG}(K_i) \in \{0, 1\}^{(\hat{H}+1)\ell}$; $\rho[a, b]$ is a byte range in units of ℓ . It checks $\gamma_i = \text{MAC}_{K_i}(\beta_i)$ and peels

$$(\beta_i \parallel 0^\ell) \oplus \rho_i = \lambda_i \parallel \beta_{i+1}.$$

The client builds, for $1 \leq i < H$, with R uniform,

$$\phi_0 = \epsilon, \quad \phi_i = (\phi_{i-1} \parallel 0^\ell) \oplus \rho_i[\hat{H}-i+1, \hat{H}+1),$$

$$\beta_H = ((\lambda_H \parallel R) \oplus \rho_H[0, \hat{H}-H+1)) \parallel \phi_{H-1},$$

$$\beta_i = (\lambda_i \parallel \beta_{i+1}[0, \hat{H}-1)) \oplus \rho_i[0, \hat{H}), \quad \gamma_i = \text{MAC}_{K_i}(\beta_i).$$

This is Sphinx's header [1]: per-hop PRG filler with MACs the client precomputes. α_{i+1} is either a per-layer envelope carried in λ_i (as above) or a blinding of α_i (Sphinx's single group element, roughly halving ℓ); L5' holds for both.

Proposition 3 (Header correctness). For $H \leq \hat{H}$ and $1 \leq i < H$, peeling β_i under K_i yields (λ_i, β_{i+1}) , peeling β_H yields λ_H , and every γ_i verifies.

Proof. By induction down from H , the last $i-1$ blocks of β_i are ϕ_{i-1} : true at H by construction, and expanding β_i and ϕ_i carries it from $i+1$ to i . So the first $\hat{H}$ blocks of $(\beta_i \parallel 0^\ell) \oplus \rho_i$ are $\lambda_i \parallel \beta_{i+1}[0, \hat{H}-1)$, and its last block $\rho_i[\hat{H}, \hat{H}+1)$ is the last block of ϕ_i , hence of β_{i+1} . $\square$

Law L3 (width). The carry slot has width $C = C_b$ on every edge of a loop of class b . With $C_0 = C - \tau$, for fixed-width symbols the client picks the least b with $|\text{enc}(v_0)| < C_0$ and $W(f_k) < C_0$ for all k ; for a stream symbol, whose width is C_0 itself, b is the client's choice between throughput and leakage (Table II). W factors through the output type ($W(f) = w(\text{Out}_f)$, so the width a consumer unpads is fixed by its own In); a **WorldFacing** stream symbol (tcp, https) has $\text{In} = \text{Out} = \text{Bytes}_{<C_0}$, one chunk per loop or reply, and larger bodies span several loops. Further,

$$\text{pad}(b) = b \parallel 80 \parallel 00^{C_0-|b|-1}, \quad |\text{enc}_f(w)| = W(f);$$

an output with $|\text{enc}_f(w)| > W(f)$ fails closed, never truncated.

Law L4 (timing). $t_i^{\text{dep}} = t_i^{\text{arr}} + L(f_i)$ exactly. An evaluation unfinished at t_i^{dep} is dropped, never released late. The one exception is a reply of a **WorldFacing** session symbol (Algorithm 2): it leaves when world bytes are ready, at any time before the SURB's x .

VI. Carry

Write E_k^τ, D_k^τ for AEZ v5 [5], [13] encryption and decryption with empty nonce and associated data; $|E_k^\tau(m)| = |m| + \tau$, and D_k^τ returns $\perp$ unless the deciphered block ends in τ zero bytes.

Algorithm 1 $\text{Hop}_i(\text{from}, \chi, y, t^{\text{arr}})$: one position, all symbols

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1:  $(\lambda, \chi') \leftarrow \text{peel}(sk_i, \chi)$ ; if  $\gamma$  fails then drop
2: if  $x \notin (t^{\text{arr}}, t^{\text{arr}} + V] \vee e \neq \text{epoch}_i \vee \nu \in R_i \vee R_i^{\text{from}}$  full then
3:   drop ▷ L9
4:  $R_i \leftarrow R_i \cup \{(\nu, t^{\text{arr}} + V)\}$ 
5: if  $f = \text{relay}$  then
6:    $y' \leftarrow D_{k_{\text{in}}}^0(y)$ 
7: else
8:    $b \leftarrow D_{k_{\text{in}}}^\tau(y)$ ; if  $b = \perp$  then drop ▷ L8
9:   if  $f$  is a WorldFacing session symbol then
10:     $Q_h \leftarrow Q_h \cup \{(\text{next}, \chi', \kappa^{\text{out}}, x)\}$  ▷ D8
11:    write  $\text{enc}^{-1}\text{pad}^{-1}(b)$  to  $\bar{a}$ 's target; return
12:    $w \leftarrow \llbracket f \rrbracket(\bar{a}, \text{enc}^{-1}\text{pad}^{-1}(b))$  by  $t^{\text{arr}} + L(f)$  ▷ L4
13:   if  $\text{late} \vee |\text{enc}_f(w)| > W(f)$  then
14:     drop ▷ L3
15:    $y' \leftarrow E_{k_1^{\text{out}}}^0 \circ \dots \circ E_{k_s^{\text{out}}}^0 \circ E_{k_{s+1}^{\text{out}}}^\tau(\text{pad enc}_f(w))$ 
16: release  $(\chi', y')$  to next at  $t^{\text{arr}} + L(f)$ 

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Definition D7 (carry). For segment k the client draws fresh keys $k_{r_{k,1}}, \dots, k_{r_{k,s}}, k_{c_k}$ and places them

$$\kappa_{p_k}^{\text{out}} = (k_{r_{k,1}}, \dots, k_{r_{k,s}}, k_{c_k}), \quad \kappa_{r_{k,j}}^{\text{in}} = k_{r_{k,j}}, \quad \kappa_{c_k}^{\text{in}} = k_{c_k}.$$

With $b_k = \text{pad}(\text{enc}(v_k))$ and $\tau = 16$:

$$y_{k,0} = E_{k_{r_{k,1}}}^0 \circ \dots \circ E_{k_{r_{k,s}}}^0 \circ E_{k_{c_k}}^\tau(b_k),$$

$$y_{k,j} = D_{k_{r_{k,j}}}^0(y_{k,j-1}), \quad b_k = D_{k_{c_k}}^\tau(y_{k,s}).$$

A value of type $()$ is encoded as ϵ , so every consumer authenticates $\text{pad}(\epsilon)$ and processing is uniform.

Since s is constant, $|\kappa_i|$ is constant and reveals no position. AEZ v5 is implemented in-repo on RustCrypto `aes::hazmat` (constant-time, `wasm32`); keys are raw 48-byte strings, a key with a zero subkey I, J or L is rejected, and $\text{GF}(2^{128})$ doubling is branch-free. $C \geq 32$ keeps every call on the AEZ-core path, never AEZ-tiny.

Proposition 4 (One value per key). Under L9 every $k_{r_{k,j}}$ and k_{c_k} is applied to exactly one C -byte value: enciphered once by p_k , deciphered once by its holder.

Proof. Keys are fresh per layer (D7). p_k enciphers its single output once under each key of $\kappa_{p_k}^{\text{out}}$. The holder of k decipheres only a cell whose layer names k , and L9 admits that layer once, across restarts by the process epoch. Without L9 a replayed cell with a modified carry makes the holder a chosen-ciphertext oracle under a fixed key: L9 is load-bearing for AEZ. $\square$

Definition 2 (Client position). $\diamond$ is position $H+1$. λ_H carries, in place of α_{H+1} , a fresh tag $t_\diamond \in \{0, 1\}^{128}$; the guard sends (χ_{H+1}, y_H) with $\chi_{H+1} = (t_\diamond, \beta_{H+1}, \gamma_{H+1})$. $\diamond$ keeps $T: t_\diamond \mapsto (k_{c_n}, \text{loop})$, removes the entry on first use, and drops a cell with an unknown tag. Every SURB suffix ends in its own tag.

Algorithm 2 $\text{Reply}_h(t_0)$: world bytes ready at session hop h

- 1: $Q_h \leftarrow \{v \in Q_h : t_0 < x_v\} \triangleright$ expired SURBs discarded
 - 2: if $Q_h = \emptyset$ then
 - 3: pause reading; return $\triangleright$ Inv. 1
 - 4: take $v = (\text{next}, \chi, k_1..k_{s+1}, x) \in Q_h$ of least x
 - 5: $w \leftarrow$ at most $C_0 - 1$ world bytes
 - 6: $y' \leftarrow E_{k_1}^0 \circ \dots \circ E_{k_s}^0 \circ E_{k_{s+1}}^\tau(\text{pad}(w))$
 - 7: release (χ, y') to next
-

$t_\diamond$ is uniform and seen only by g at position H and by $\diamond$, so it links nothing the guard cannot already link by timing.

VII. Reply Blocks

Definition D8 (SURB). A single-use reply block for a session symbol at h is the client-sealed loop suffix after h :

$$v = (\text{next}, \chi_v, k_{r_1}, \dots, k_{r_s}, k_\diamond, x_v).$$

h keeps the suffix of every forward loop that reaches it $(\chi_{i+1}, \kappa_i^{\text{out}}, x)$ as one SURB; no SURB rides in a carry value.

Proposition 5 (No SURB in the carry). A carried SURB forces $C > \hat{H}\ell$ on every edge.

Proof. L5' fixes $|\chi_v| = |\chi| > |\beta| = \hat{H}\ell$ (a shorter reply header would name the reply path), and L3 fixes one C for every edge of a loop and its replies. $\square$

At $s = 2$ that would at least double every cell (9086 B of β in the carry besides the header's own), so credit arrives only as forward loops: one forward cell per reply cell, data optional.

Invariant 1 (Credit). Let Q_h be h 's unexpired SURBs (t_v the arrival of v), $u(t)$ the forward loops of the session received and $r(t)$ the replies emitted by time t . Then

$$r(t) \leq u(t), \quad v \in Q_h \Rightarrow t < x_v \leq t_v + V, \\ |Q_h| = 0 \Rightarrow \text{tcp stops reading its socket.}$$

Each reply consumes one v , produced as segment n of D7 with $p_n = h$; an expired v is discarded unused; a new forward loop resumes reading. By the rule for x , a SURB lives at most $X_0 + Q = V$, and a session idle for longer than V holds no credit.

VIII. Security Laws

Assumption 1 (Primitives). Under a key it does not hold, a hop sees E_k^0 as a uniformly random permutation of $\{0, 1\}^C$ and E_k^τ as robust AE [5]; PRG and MAC are secure, and key transport reveals K_j to hop j only. Network delays δ are fixed per edge; no hop other than g observes the client's departure time.

Definition D9 (view). Over client coins ρ (keys, nonces, header randomness) and delays δ ,

$$\text{view}_i = (f_i, \bar{a}_i, v_{i-1}, \text{from}_i, \text{next}_i, e_i, x, b, \\ t_i^{\text{arr}}, t_i^{\text{dep}}, \chi_i, y_{i-1}, \chi_{i+1}, y_i),$$

with $v_{i-1} = \perp$ when $f_i = \text{relay}$ (a relay consumes no value).

Law L5 (noninterference). For loops P, P' let

$$\pi_i(P) = (f_i, \bar{a}_i, v_{i-1}, \text{from}_i, \text{next}_i, e_i, x, b), \\ \sigma(P) = f_1 \cdots f_H, \\ \pi_i(P) = \pi_i(P') \wedge \sigma(P) = \sigma(P') \\ \Rightarrow \text{view}_i(P) \approx_c \text{view}_i(P').$$

Symbols may leak; $\bar{a}_j, v_j$ for $j \neq i$, other than the consumed v_{i-1} , may not [14].

Proposition 6 (L5 from D6, D7, L3, L4). Under Assumption 1, every loop built by D4c–D7 satisfies L5. Equality in distribution holds for the carry alone; the header makes L5 computational (Prop. 7).

Proof. Lengths: $|\chi_i|$ and $|y_i| = C$ are constant (D6, L3). Times: $t_i^{\text{arr}} = \sum_{j < i} (L(f_j) + \delta_j)$ by L4, a function of σ, δ . Headers: χ_i, χ_{i+1} reveal only λ_i , fixed by π_i up to fresh coins (Prop. 7). Carry: if $i = r_{k,j}$, then $D_{k_{r_{k,j}}}^0(y_{k,j-1})$ is an E output under $k_{r_{k,j+1}}$ or k_{c_k} , a fresh key not held by i , hence uniform on $\{0, 1\}^C$ and independent of v_k . If $i = c_k$, it learns v_{i-1} , which π_i fixes. If $i = p_k$, its output is computed from v_{i-1} and $\bar{a}_i$. No component depends on $\bar{a}_j, v_j$ otherwise. $\square$

L5's premise fixes σ , hence H and every index, so it says nothing about length or position; L5' does.

Law L5' (position hiding). For loops P, P' with $H(P), H(P') \leq \hat{H}$ and positions i of P , i' of P' ,

$$\pi_i^\circ(P) = (f_i, \bar{a}_i, v_{i-1}, \text{from}_i, \text{next}_i, b), \\ \pi_i^\circ(P) = \pi_{i'}^\circ(P') \Rightarrow \text{view}_i(P) \approx_c \text{view}_{i'}(P').$$

No hop learns its position, H , or the symbols elsewhere in the loop, except the guard: $\text{from} = \diamond$ or $\text{next} = \diamond$ names position 1 or H , and π° records it.

Proposition 7 (L5' from Def. 1). Under Assumption 1, the distinguishing advantage is at most $\hat{H} \epsilon_{\text{PRG}} + \epsilon_{\text{MAC}} + \epsilon_\alpha$.

Proof. $|\chi|$ and C do not depend on H or i . e_i is the hop's own, x depends only on the build quantum (D6), and t_i^{arr} carries no position because the client's departure time is unobserved (Assumption 1). The carry is uniform at a relay position (Prop. 6) and fixed by $v_{i-1} \in \pi_i^\circ$ at a consumer. Hop i sees λ_i , fixed by π_i° up to fresh coins, α_{i+1} , γ_{i+1} , and β_{i+1} . Every block of β_{i+1} but the last is masked by some ρ_j , $j \neq i$: by ρ_{i+1} when $i+1 < H$, and by ρ_H or an upstream term of ϕ_{H-1} when $i+1 = H$. The last block is $\rho_i(\hat{H}, \hat{H}+1)$, computed by i itself in both loops. Hybrids replacing the ρ_j , $j \neq i$, one key at a time by uniform strings make β_{i+1} uniform apart from that block,

in P and P' alike; γ_{i+1} is then a MAC under an unseen key. $\square$

Law L6 (path authority). next_i is a field of λ_i , fixed by $\diamond$ at construction: next_i is independent of every v_j . No symbol chooses a hop.

Proposition 8 (Admissible combinators). L6 admits the composition fragment ($\ggg$, without *arr* or *first*) of Arrows [9] and excludes monadic bind. Selective branching [11] over client-sealed branches needs L6 weakened to $\text{next}_i \in N_i$, a finite set sealed by $\diamond$.

Proof. $\ggg$ fixes every hop before any value exists. *select* : $F(A + B) \rightarrow F(A \rightarrow B) \rightarrow F B$ fixes the possible effects, so $\diamond$ can seal both branches, but the selecting hop picks next_i by v_i and learns the branch taken; its law is deferred with Selective itself. Bind $m \ggg k$, $k : A \rightarrow M B$, chooses the continuation, and so its hops, from a run-time value: either a hop constructs next (violating L6) or $\diamond$ seals $|A|$ continuations. This is the arrows-meticulous, monads-promiscuous separation of [10]. $\square$

“OnionMonad” names M , the effect inside one hop; across hops only the static fragment of $\mathbf{KI}(M)$ is exposed.

Law L8 (unlinkability). On consecutive edges $\chi_i \neq \chi_{i+1}$ and $y_{i-1} \neq y_i$ (one layer peeled, one AEZ layer removed per hop). If any $y_{k,j}$ is modified to $\tilde{y} \neq y_{k,j}$,

$$\Pr[D_{kc_k}^\tau(\tilde{y}_{k,s}) \neq \perp] \leq 2^{-8\tau} + \epsilon_{\text{AEZ}},$$

and a rejected value yields nothing; no bit of a tag survives. L8 is a byte-level law; colluders still link by timing (Table II).

Under Assumption 1 consecutive y are independent uniform strings to any hop lacking both keys. A modification propagates through every later wide-block decipherment, so, unlike counter-mode relay cryptography [15], a flipped bit is never observed in place; this is Sphinx’s wide-block payload [1], [16] with AEZ in place of LIONESS.

Law L9 (replay). With $V = 150\text{ s}$ (ONION_FORWARD_MAX_VALIDITY_MS) and per-sender capacity $B = 4096$ over at most 64 senders, hop i admits a layer at t^{arr} iff

$$t^{\text{arr}} < x \leq t^{\text{arr}} + V \wedge \nu \notin R_i \wedge |R_i^{\text{from}}| < B,$$

then retains ν in R_i until $t^{\text{arr}} + V$. A full R_i^{from} rejects; no tag leaves R_i before its retention ends. A sender beyond the 64th takes the quota of the least recently active partition, whose tags stay in R_i until their retention ends; admission rejects while $|R_i| = 64B$. A SURB is a layer, hence single use; L9 also bounds each AEZ key to one carry value (Prop. 4).

Proposition 9 (Replay bounds). (i) No admitted layer outlives its tag: $x \leq t^{\text{arr}} + V$. (ii) $|R_i| \leq 64B$. (iii) Hop i admits at most B layers per sender per V , i.e. $B/V \approx 27$ cells s^{-1} per neighbour. (iv) A SURB is usable only within $[x_v - V, x_v)$, so SURB lifetime is at most V . (v) Hop i admits at most $64B/V \approx 1748$ cells s^{-1} in total.

Table II
Residual leakage, stated and accepted

leakage	bound
symbols σ are public; λ_i names f_i	L5 hypothesis
a WorldFacing hop knows it faces the world	no other hop learns whether symbols follow (L3, D6, L5')
g sees departure and return	round-trip $\sum_e d_e + \sum_i L(f_i)$: an estimate of H and of $\sum_i L(f_i)$; normalising it is a separate issue
latency class	L4 quantises timing; cannot remove it
hosted backend	sees $\bar{a}$ of every application of its symbol
loop class b	every hop sees it; the same leakage as today’s cell buckets
session reply timing	follows the world, not $L(f)$ (L4); a session hop may hold a SURB until its x
stream class b	chosen by the client for a stream symbol; reveals the throughput class
expiry x	shared by all hops of a loop and all loops sealed in one $Q = 30\text{ s}$ quantum: colluders link positions only within that set
p_k holds $\kappa_{p_k}^{\text{out}}$	p_k colluding with $r_{k,j}$ links two positions of one segment; no value leaks, and timing links them anyway
client-authored semantics ($\bar{a}_{i+1}$ depends on meaning of v_i)	out of protocol scope

Proof. (i) is the admission window; (ii) the capacity guard; (iii) each admitted tag occupies a slot for V ; (iv) every relay of the reply path applies (i) at the reply’s arrival; (v) follows from (ii). $\square$

(iii) bounds per-neighbour loop throughput, including SURB replies; B is a tuning constant, not a law.

Law L10 (wire). Phase 1 preserves today’s bytes (golden tests); Phase 2 replaces them with a new pinned golden set in one cutover, without versioning.

IX. Accepted Leakage

Table II is exhaustive with respect to L5: every other component of view_i is determined by π_i and σ .

X. The Current Design

Proposition 10 (Backward linkability). Today the exit seals its answer once, $c = \text{AEAD}_{p_k}(\cdot)$ (`encrypt_client_payload_at_sequence` in `circuit/crypto.rs`), and each relay re-emits it as *payload* = *frame.payload* (`advance_backward` in `circuit/reducer.rs`), rewriting only the edge circuit id and the outer hop seal. Every backward relay therefore sees the same c : any two colluding relays, and the exit, link their positions by equality, independently of route length.

Table III
Current circuit versus OnionMonad

concern	current	here
layers	Relay, Exit	one uniform λ_i (D6)
header	nested, -404B per hop	fixed $\hat{H}\ell$ (L5')
length	3 by default, ≤ 8 , short paths allowed	$H(n, 2) \leq 14$ (D5)
computation	exit service only	any $f \in \Sigma_n$ (D2)
return path	forward path reversed	fresh relays to g (L7)
relay state	relay_returns, ≤ 1024 , TTL 120s refreshed per cell	replay tags only (L9)
return bytes	one ciphertext, identical per edge	one AEZ layer per edge (L8)
replay	exits (consume_forward_nonce), $V = 150\text{s}$, $B = 4096$	every hop, same V , B (L9)
session replies	no SURB credit	one per SURB (Inv. 1)

Proposition 11 (Nested layers leak position). `build_forward_layers_with_ids` nests one envelope per hop, so hop i decrypts $inner_i$ with $|inner_i| = \sum_{j>i} \ell_j + |\lambda_H|$, measured at 404B per downstream hop. A loop extended after i changes $|inner_i|$, so $view_i$ distinguishes it with probability 1: L5' fails.

Cell buckets pad the cell before decryption, not the layer; padding the innermost layer to a constant total only moves the leak from $H - i$ to i .

The stored return edge $(id', \text{next}) \mapsto (\text{from}, id, pk)$ is the only reason relays hold circuit state; the loop removes it with Backward, OnionBackwardFrame, MAX_ONION_RELAY_CIRCUITS, and ONION_RELAY_RETURN_TTL_MS.

XI. Security

Table IV maps each threat to the law that bounds it.

The adversary controls any set of nodes other than $\diamond$, and delivery order; it cannot break Assumption 1. Relays do not mix, so this is Tor's threat model: a global passive observer is out of scope. Dropping is always possible and out of scope.

XII. Verification

Every law is discharged against the pure builder and reducer, not a shadow model (Table V).

XIII. Phases

Table VI orders the cutover. Fixed: $s = 2$, $n_{\max} = 4$, $\hat{H} = 14$, in-repo AEZ. Open: key transport (per-layer envelope or one blinded element), raw DID and key encodings ($\ell(2, 64) \approx 531\text{B}$), and the replay capacity B . Non-goals: bind, Selective branches before Phase 2b, session symbols in intermediate position, concrete symbols, and verification of results.

Table IV
Threats and the law that bounds each

threat	bound
hop learns	L5 (Prop. 6)
another's $\bar{a}_j, v_j$	
position, loop length	fixed $ \beta $, C , $ \kappa $, s (D5–D7, L3, L5')
value-dependent timing	exact release $t^{\text{arr}} + L(f)$ (L4)
colluding relays	per-edge re-encipherment (L8), up to the shared x (Table II); cf. Prop. 10
match bytes	whole-value rejection, $2^{-8\tau}$ (L8)
tagging	$\text{next}_i \in \lambda_i$ (L6); bind excluded (Prop. 8)
hop-chosen routing	one guard closes the loop (L7)
first/last-hop correlation	
replay of layer or SURB	single admission per ν (L9)
replay across a restart	new process epoch rejects every older layer (D2, L9)
reply	$r(t) \leq u(t)$ (Inv. 1)
amplification	
symbol abuse	per-symbol quota(f) (D1)
oversized result	fail closed at $W(f)$ (L3)

XIV. Related Work

Tor [12] answers along the circuit it built, as Rings does today; Degabriele and Stam [15] formalise the tagging that its counter-mode relay cryptography admits. Chaum's untraceable return addresses [2] and Mixminion's SURBs [3] are the source of D8; Sphinx [1] gives the fixed-length header and a LIONESS [16] payload whose zero-block check L8 inherits. Loopix [4] routes traffic as loops for cover; here the loop is the circuit. AEZ [5], [13] supplies both the length-preserving relay layer and robust AE in one primitive. On the semantic side, Moggi's Kleisli semantics [6], generic effects and handlers [7], [8], and the Arrow/Selective/Monad hierarchy [9]–[11] fix which compositions a client can seal in advance.

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Table V
Obligations and the test that discharges each

obligation	test or property
L10	golden encodings of layers, wire messages, cells, and descriptor signing data, landed before any refactor; a new pinned set at the Phase 2 cutover
L1	carry through <code>relay</code> is identity; $\text{Loop}(\text{relay} \gg P) \stackrel{d}{=} \text{Loop}(P)$
D4a	<code>Out</code> $\neq$ <code>In</code> , <code>WorldFacing</code> before the last symbol, empty pipeline: rejected at construction
L3	fixed slot width on every edge; output $> W(f)$ dropped
L4	release at $t^{\text{arr}} + L(f)$ and drop on overrun under an injected clock
L5	property test on the pure builder: vary $\bar{a}_j, v_j, j \neq i$; view _{<i>i</i>} bytes, lengths, and release times equal
L5'	builder for $H \neq H' \leq \hat{H}$: equal layer and cell lengths at every position; every γ_i verifies (Prop. 3)
L3, D6	one class b and slot width C_b on every edge of a loop, SURB replies included; stream chunks $< C_0$
client	reply accepted once by tag $t_\circ$; unknown or reused tag dropped
D5, L7	$H = (s+1)n + s$; guard at 1 and H only; $n > 4$ rejected; selection fails closed below $H - 1$ eligible DIDs
L8	per-hop byte inequality; one-bit flip rejects the whole value
L9	replayed layer and reused SURB rejected; a layer for an earlier process epoch rejected after restart; $x \notin (t^{\text{arr}}, t^{\text{arr}} + V]$ rejected; full R_i^{from} rejects without evicting a live tag
credit	a tcp session emits no reply without a SURB, drops expired SURBs, and resumes after a new forward loop
AEZ	2941 vectors from the AEZ v5 reference C code (length sweep, parameter grid, CAESAR genkat), bit-flip rejection for $\tau \geq 4$, native and wasm32; weak-key rejection; fuzz/ target for $D \circ E = \text{id}$ and panic freedom

Table VI
Cutover plan

phase	content
1	golden bytes; closed Σ table (unknown names rejected, no tcp fallback; configs change, bytes do not); pipeline term; layer build as a fold over P ; reducer and handler as Σ -algebra; route by symbol sequence. Zero wire change, no bump
2a	uniform layer, fixed-length header ($L5'$), loop with derived $\hat{H}$, loop classes, relay process epoch, client tag, stateless relays, AEZ carry, SURBs. Breaking
2b	intermediate symbols: Invoke with release time, quota(f). Same cutover; one version bump on its last PR

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